

Part I

Recitation Questions

Problem 1

February 21, 2026

Problem 1

Problem 1 (a) Let $f(x) = \frac{x-1}{x^2-5x}$. Then $f(2) = -\frac{1}{6}$ and $f(6) = \frac{5}{6}$, but there is no value of c between 2 and 6 for which $f(c) = 0$. Does this fact violate the Intermediate Value Theorem?

Explanation. It does not violate the Intermediate Value Theorem. f is not continuous at 5 so the conditions of the IVT do not hold and therefore the IVT does not apply.

(b) True or False: At some time since you were born your weight in pounds exactly equaled your height in inches.

Explanation. True: if $w(t)$ represents your weight in pounds at time t and $h(t)$ represents your height in inches at time t , then w and h are both continuous functions. This implies $w - h$ is also continuous. If $t = 0$ is the moment you were born and $t = T_0$ is the present time, then $w(0) - h(0) < 0$ and $w(T_0) - h(T_0) > 0$. Hence by the Intermediate Value Theorem there is a point in the past, t , when $w(t) - h(t) = 0$ and therefore your weight in pounds equaled your height in inches.

Problem 2

February 21, 2026

Problem 2 For the following function g defined by

$$g(t) = \begin{cases} 5t + 7 & \text{if } t < -3 \\ \frac{(t-1)(t+2)}{t+2} & \text{if } -3 \leq t < 1 \text{ and } t \neq -2 \\ 4 \ln t & \text{if } t \geq 1 \end{cases}$$

find the **Intervals of Continuity**.

(**Important Note:** Write your answer as a list of intervals, with each interval separated by a comma.)

Explanation. Recall that *Intervals of Continuity* means the largest such intervals on which the function is continuous.

The function g is continuous on $(-\infty, -3)$ since, in this interval, $g(t) = 5t + 7$ is a polynomial and therefore continuous on its domain.

g is not continuous at $t = -3$:

$$\lim_{t \rightarrow -3^-} g(t) = \lim_{t \rightarrow -3^-} (5t + 7) = 5(-3) + 7 = -8$$

and

$$\lim_{t \rightarrow -3^+} g(t) = g(-3) = \frac{(-3-1)(-3+2)}{-3+2} = \frac{4}{-1} = -4.$$

Although we have found g is not continuous at $t = -3$, since $\lim_{t \rightarrow -3^+} g(t) = g(-3)$, g is continuous from the right at $t = -3$.

The function g is continuous on $[-3, -2)$, $(-2, 1)$ since, in this interval, $g(t) = \frac{(t-1)(t+2)}{t+2}$ is a rational function whose denominator is not zero. g is not continuous at $t = -2$ because it is not defined there. g is continuous on $(1, \infty)$, since, on this interval, $g(t) = 4 \ln(t)$ is a constant multiple of a logarithmic function.

g is continuous at $t = 1$:

$$\lim_{t \rightarrow 1^-} g(t) = \lim_{t \rightarrow 1^-} \frac{(t-1)(t+2)}{t+2} = \frac{0}{3} = 0,$$

and

$$\lim_{t \rightarrow 1^+} g(t) = g(1) = 4 \ln(1) = 0$$

So, the Intervals of Continuity are

$$(-\infty, -3), [-3, -2), (-2, \infty)$$

Problem 3

February 21, 2026

Problem 3 Determine the value of a constant b for which f is continuous at 0. **EXPLAIN.**

$$f(x) = \begin{cases} \frac{2x+b}{x-5} & \text{if } x < 0 \\ \frac{x+16}{x^2-16} & \text{if } x \geq 0 \end{cases}$$

Explanation. In order for f to be continuous at $x = 0$, the three conditions of continuity have to be satisfied. These criteria are:

$$\begin{aligned} &f \text{ is defined at } x = 0 \\ &\lim_{x \rightarrow 0} f(x) \text{ exists} \\ &\lim_{x \rightarrow 0} f(x) = f(0) \end{aligned}$$

The first condition is satisfied. In order to satisfy the second condition, $\lim_{x \rightarrow 0} f(x)$ exists, we must verify that $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$. That is,

$$\begin{aligned} \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} \left(\frac{x+16}{x^2-16} \right) = \frac{16}{-16} = -1 \\ \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} \frac{2x+b}{x-5} = \frac{b}{-5} \\ \implies \lim_{x \rightarrow 0} f(x) &= -1 = \frac{-b}{5} \implies b = 5 \end{aligned}$$

and $f(0) = -1$. f is continuous at $x = 0$ for $b = 5$ since $\lim_{x \rightarrow 0} f(x) = f(0) = -1$.

Notice here, it's not enough to just find a value for b . Our work must also justify the continuity at $x = 0$ for that value of b .

Problem 4

February 21, 2026

Problem 4

Problem 4 Use the Intermediate value theorem to find an interval in which you can guarantee that there is a solution to the equation $x^3 = x + \sin(x) + 1$. **EXPLAIN.** (Do not use a graphing device or calculator to solve this problem!)

Explanation. We have

$$x^3 = x + \sin(x) + 1 \iff x^3 - x - \sin(x) - 1 = 0.$$

Define $f(x) = x^3 - x - \sin(x) - 1$.

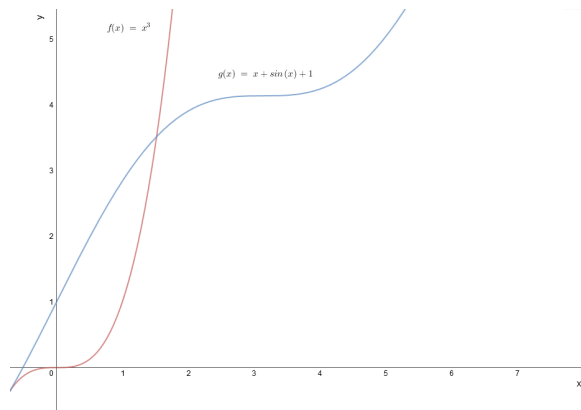
Since:

$$f(0) = (0)^3 - (0) - \sin(0) - 1 = -1 \text{ and}$$

$$f(\pi) = \pi^3 - \pi - \sin(\pi) - 1 = \pi(\pi^2 - 1) - 1 > 3 \cdot (3^2 - 1) - 1 = 23$$

We have: $-1 < 0 < 23$

and f is continuous on $[0, \pi]$, the Intermediate Value Theorem implies there is some c with $0 < c < \pi$ such that $f(c) = 0$, that is, $c^3 = c + \sin(c) + 1$. That c is a solution on the interval $[0, \pi]$. (Note: There are many possible intervals, e.g. $[1, 4]$, etc.)



Problem 5

February 21, 2026

Problem 5 (a) True or False: If f and g are two functions defined on $(-1, 1)$, and if $\lim_{x \rightarrow 0} g(x) = 0$, then it must be true that $\lim_{x \rightarrow 0} (f(x) \cdot g(x)) = 0$.

Explanation. False: Suppose

$$f(x) = \begin{cases} \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

and $g(x) = x$. Then $\lim_{x \rightarrow 0} g(x) = 0$ but

$$\lim_{x \rightarrow 0} (f(x) \cdot g(x)) = \lim_{x \rightarrow 0} \left(\frac{1}{x} \cdot x \right) = \lim_{x \rightarrow 0} 1 = 1.$$

(b) True or False: If f is continuous on $(-1, 1)$, and if $f(0) = 10$ and $\lim_{x \rightarrow 0} g(x) = 2$, then

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = 5.$$

Explanation. True: application of quotient rule. Because f is continuous, $\lim_{x \rightarrow 0} f(x) = f(0) = 10$

(c) True or False: If f is continuous on $[1, 3]$, and if $f(1) = 0$ and $f(3) = 4$, then the equation $f(x) = \pi$ has a solution in $(1, 3)$.

Explanation. True: f is continuous on $[1, 3]$, $f(1) = 0 < \pi < 4 = f(3)$, and the Intermediate Value Theorem implies there is some x in $(1, 3)$ with $f(x) = \pi$.

(d) True or False: Let f be a positive function with vertical asymptote $x = 5$. Then

$$\lim_{x \rightarrow 5} f(x) = \infty.$$

Explanation. False: Suppose f is defined by

$$f(x) = \begin{cases} \frac{1}{x-5} & \text{if } x > 5 \\ 2 & \text{if } x \leq 5. \end{cases}$$

Then f has a vertical asymptote at $x = 5$:

$$\lim_{x \rightarrow 5^+} \underbrace{\frac{1}{x-5}} = \infty,$$

because the limit is of the form $\frac{\neq}{0}$, the numerator positive, and the denominator positive and goes to 0.

But, $\lim_{x \rightarrow 5^-} f(x) = 2$.

Therefore, $\lim_{x \rightarrow 5} f(x)$ does not exist

Problem 6

February 21, 2026

Problem 6 Let

$$h(u) = \begin{cases} \frac{u^2 - 5u + 4}{u - 4} & \text{if } u < 4 \\ \frac{-\sqrt{u+4}}{u-6} & \text{if } u \geq 4, u \neq 6. \end{cases}$$

(a) What is the domain of h ?

Explanation. For values of $u < 4$ we check $\frac{u^2 - 5u + 4}{u - 4}$. This piece of the function is defined everywhere on $(-\infty, 4)$

For values of $u \geq 4$ we check $\frac{-\sqrt{u+4}}{u-6}$. This piece of the function is defined everywhere except $u = 6$. So h is defined on $[4, 6) \cup (6, \infty)$. The domain is $(-\infty, 6) \cup (6, \infty)$

(b) Find all vertical asymptotes of h . **EXPLAIN.**

Explanation. The only candidates are $u = 6$ and $u = 4$.

$$\lim_{u \rightarrow 6^+} \frac{-\sqrt{u+4}}{u-6} = -\infty \text{ because it is of the form } \frac{\#}{0}, \text{ the numerator negative, the denominator positive and}$$

$$\lim_{u \rightarrow 4^-} \frac{u^2 - 5u + 4}{u - 4} = \lim_{u \rightarrow 4^-} \frac{(u-1)(u-4)}{u-4} = \lim_{u \rightarrow 4^-} (u-1) = 3$$

$$\lim_{u \rightarrow 4^+} \frac{-\sqrt{u+4}}{u-6} = \frac{-\sqrt{8}}{-2} = \sqrt{2}$$

The only vertical asymptote is $u = 6$, because $\lim_{u \rightarrow 6^+} h(u) = -\infty$.

(c) Find all horizontal asymptotes of h . **EXPLAIN.**

Explanation.

$$\begin{aligned}\lim_{u \rightarrow \infty} \frac{-\sqrt{u+4}}{u-6} &= \lim_{u \rightarrow \infty} \frac{-\frac{\sqrt{u+4}}{\sqrt{u}}}{\frac{u-6}{\sqrt{u}}} \\ &= \lim_{u \rightarrow \infty} \frac{-\sqrt{\frac{u+4}{u}}}{\frac{u}{\sqrt{u}} - \frac{6}{\sqrt{u}}} \\ &= \lim_{u \rightarrow \infty} \frac{\sqrt{1 + \frac{4}{u}}}{\sqrt{u} - \frac{6}{\sqrt{u}}} \\ &= 0\end{aligned}$$

because it is of the form: $\frac{\#}{\infty}$

This indicates that $y = 0$ is a horizontal asymptote.

Checking as u approaches $-\infty$ we have:

$$\lim_{u \rightarrow -\infty} \frac{u^2 - 5u + 4}{u - 4} = \lim_{u \rightarrow -\infty} \frac{(u-1)(u-4)}{u-4} = \lim_{u \rightarrow -\infty} (u-1) = -\infty$$

Therefore, there is no horizontal asymptote as u approaches $-\infty$

There is one horizontal asymptote at $y = 0$, because $\lim_{u \rightarrow \infty} h(u) = 0$.

(d) List the **Intervals of Continuity** for the function h .

Explanation. Remember that finding Intervals of Continuity means to find the largest such intervals. For values of $u < 4$ we check $\frac{u^2 - 5u + 4}{u - 4}$. This is a piece of a rational function and is continuous everywhere on $(-\infty, 4)$.

For values of $u > 4$ we check $\frac{-\sqrt{u+4}}{u-6}$. This piece of the function is continuous everywhere where defined: $(4, 6), (6, \infty)$

We need to check $u = 4$ for continuity. A function is continuous at a point a if:

$$\begin{aligned}f &\text{ is defined at } x = a \\ \lim_{x \rightarrow a} f(x) &\text{ exists} \\ \lim_{x \rightarrow a} f(x) &= f(a)\end{aligned}$$

Problem 6

h is defined at $u = 4$, $h(4) = \frac{-\sqrt{4+4}}{4-6} = \sqrt{2}$

Next we check if $\lim_{u \rightarrow 4} h(u)$ exists. That is, does $\lim_{u \rightarrow 4^-} h(u) = \lim_{u \rightarrow 4^+} h(u)$?

$$\lim_{u \rightarrow 4^-} h(u) = \lim_{u \rightarrow 4^-} \frac{u^2 - 5u + 4}{u - 4}$$

This is of the form: $\frac{0}{0^-}$ so we need to do more algebra

$$\begin{aligned} &= \lim_{u \rightarrow 4^-} \frac{(u-4)(u-1)}{(u-4)} \\ &= \lim_{u \rightarrow 4^-} (u-1) \\ &= 4-1 \\ &= 3 \end{aligned}$$

$$\begin{aligned} \lim_{u \rightarrow 4^+} h(u) &= \lim_{u \rightarrow 4^+} \frac{-\sqrt{u+4}}{u-6} \\ &= \frac{-\sqrt{4+4}}{4-6} \\ &= \frac{-\sqrt{8}}{-2} \\ &= \sqrt{2} \end{aligned}$$

$$\lim_{u \rightarrow 4^-} h(u) \neq \lim_{u \rightarrow 4^+} h(u) \implies \lim_{u \rightarrow 4} h(u) \text{ does not exist}$$

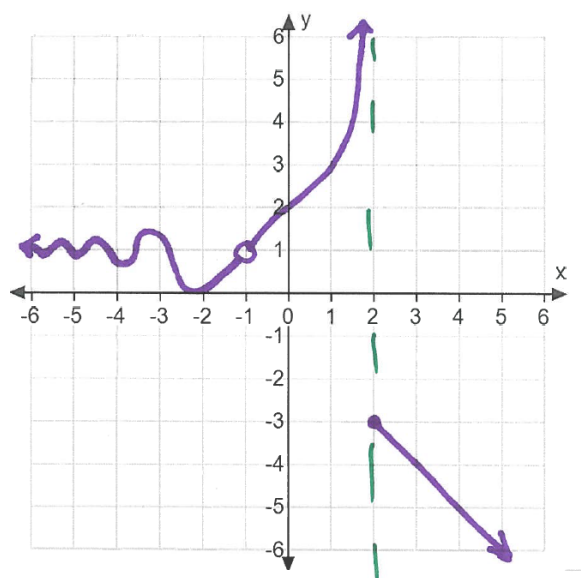
Thus $h(u)$ is not continuous at $u = 4$. Remark: Since $\lim_{u \rightarrow 4^+} h(u) = h(4)$, h is right continuous at $u = 4$. Its intervals of continuity are $(-\infty, 4)$, $[4, 6)$, $(6, \infty)$

Problem 7

February 21, 2026

Problem 7

Problem 7 Use the graph of f to answer the questions below.



(a) State the domain of f .

Explanation. $(-\infty, -1) \cup (-1, \infty)$

(b) Find the following values or state "does not exist":

(i) $\lim_{x \rightarrow 2^-} f(x) =$

(ii) $\lim_{x \rightarrow 2^+} f(x) =$

(iii) $\lim_{x \rightarrow 2} f(x) =$

(iv) $\lim_{x \rightarrow -1} f(x) =$

(v) $\lim_{x \rightarrow 3} f(x) =$

(vi) $\lim_{x \rightarrow -\infty} f(x) =$

(vii) $f(-1) =$

Explanation. (i) $\lim_{x \rightarrow 2^-} f(x) = \infty$

(ii) $\lim_{x \rightarrow 2^+} f(x) = -3$

(iii) $\lim_{x \rightarrow 2} f(x)$ Does not exist

(iv) $\lim_{x \rightarrow -1} f(x) = 1$

(v) $\lim_{x \rightarrow 3} f(x) = -4$

Problem 7

(vi) $\lim_{x \rightarrow -\infty} f(x) = 1$

(vii) $f(-1)$ Does not exist

(c) *State the equation of any vertical asymptotes.*

Explanation. $x = 2$

(d) *State the equation of any horizontal asymptotes.*

Explanation. $y = 1$

(e) *Find the Intervals of Continuity.*

Explanation. $(-\infty, -1), (-1, 2), [2, \infty)$

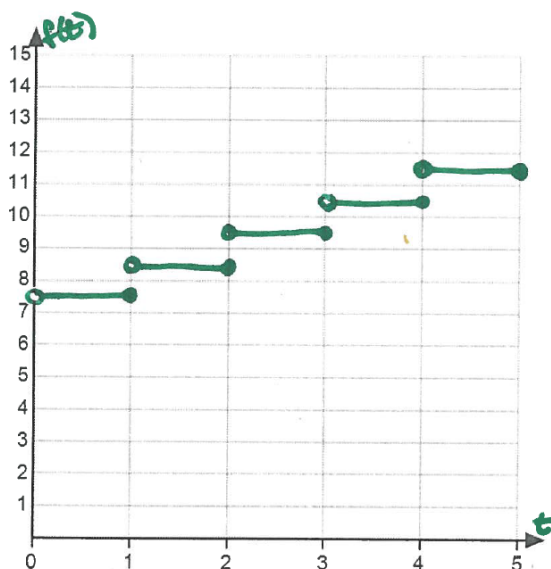
Problem 8

February 21, 2026

Problem 8 Suppose a taxi ride costs \$7.50 for the first mile (or any part of the first mile), plus an additional \$1.00 for each additional mile (or any part of a mile).

- (a) Graph the function $c = f(t)$ that gives the cost of a taxi ride for t miles, for $0 \leq t \leq 5$.

Explanation.



- (b) Evaluate $\lim_{t \rightarrow 2.9} f(t)$

Explanation. $\lim_{t \rightarrow 2.9} f(t) = 9.5$

- (c) Evaluate $\lim_{t \rightarrow 3^-} f(t)$ and $\lim_{t \rightarrow 3^+} f(t)$

Explanation. $\lim_{t \rightarrow 3^-} f(t) = 9.5$ and $\lim_{t \rightarrow 3^+} f(t) = 10.5$

- (d) Interpret the meaning of the limits in part (c).

Explanation. As the number of miles the taxi drives approaches 3, the cost of the taxi ride is \$9.50. If one drives for a bit more than 3 miles, the cost is \$10.50.

- (e) On what intervals is the function c continuous? Explain.

Explanation. $(0, 1], (1, 2], (2, 3], (3, 4], (4, 5]$

Problem 9

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Problem 9 (a) Given function f on an interval $[a, b]$:

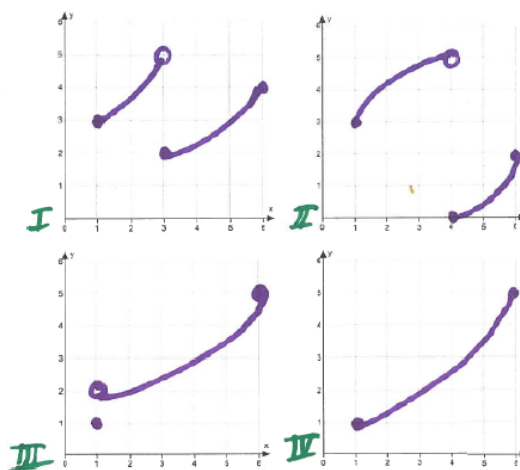
(i) What are the conditions of the Intermediate Value Theorem?

Explanation. f is continuous on the interval $[a, b]$

(ii) What is the conclusion of the Intermediate Value Theorem?

Explanation. For any number L strictly between $f(a)$ and $f(b)$, there is at least one number c in (a, b) satisfying $f(c) = L$. This means that for any L strictly between $f(a)$ and $f(b)$, the horizontal line $y = L$ intersects the graph of f .

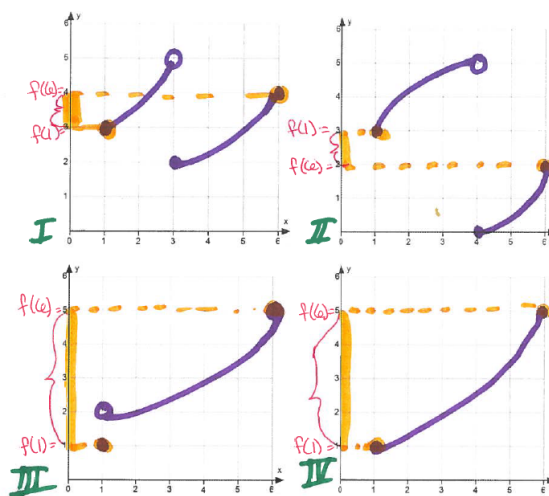
(b) Given the four functions on the interval $[1, 6]$, answer the questions below.



(i) For each of the functions I through IV, indicate $f(1)$ and $f(6)$. Then mark the interval of all numbers strictly between $f(1)$ and $f(6)$, on the y-axis.

Explanation.

Problem 9



- (ii) For each of the functions I through IV, write an interval of all numbers strictly between $f(1)$ and $f(6)$

Explanation. $f = I$: $(3, 4)$

$f = II$: $(2, 3)$

$f = III$: $(1, 5)$

$f = IV$: $(1, 5)$

- (iii) List the functions that satisfy the conditions of the Intermediate Value Theorem on $[1, 6]$

Explanation. Only function IV is continuous on $[1, 6]$

- (iv) For which of the functions is the following statement true: For any number L strictly between $f(1)$ and $f(6)$, there exists a number c in $(1, 6)$ satisfying $f(c) = L$.

Explanation. The statement is true for functions I and IV

- (v) Does the function III satisfy the conclusion of the Intermediate Value Theorem? Why or why not?

Explanation. No, function III does not satisfy the conclusion of the IVT. For example, take $L = 1.5$. $f(c) = 1.5$ has no solution in $(1, 6)$. Draw the horizontal line $y = 1.5$. What do you notice? This line does not intersect the graph of function III. What does this mean? The function does not attain that value, 1.5. So there is no c in $(1, 6)$ such that $f(c) = 1.5$.